LoRA: unseen-match questions

Research question: Can LoRA adapt the pretrained model to the held-out questions while updating fewer parameters than full fine-tuning?

Data and evaluation are taken directly from the supplied report files. This report does not retrain either model. LoRA report: C:\Users\owenv\Documents\Projects\MQP\MQP-Repo\transformer_syed\minGPT\out\lora-sports-fp16\report.json. Full-training report: C:\Users\owenv\Documents\Projects\MQP\MQP-Repo\transformer_syed\minGPT\out\full-sports-fp16\report.json.

Protocol: LoRA recorded FP16; LoRA steps 300; full-training steps 10. LoRA configuration: batch 4, length 64, rank 4, alpha 8.

Held-out answer NLL: pretrained base 2.9955 -> LoRA 0.0042; full fine-tuning 2.0696.

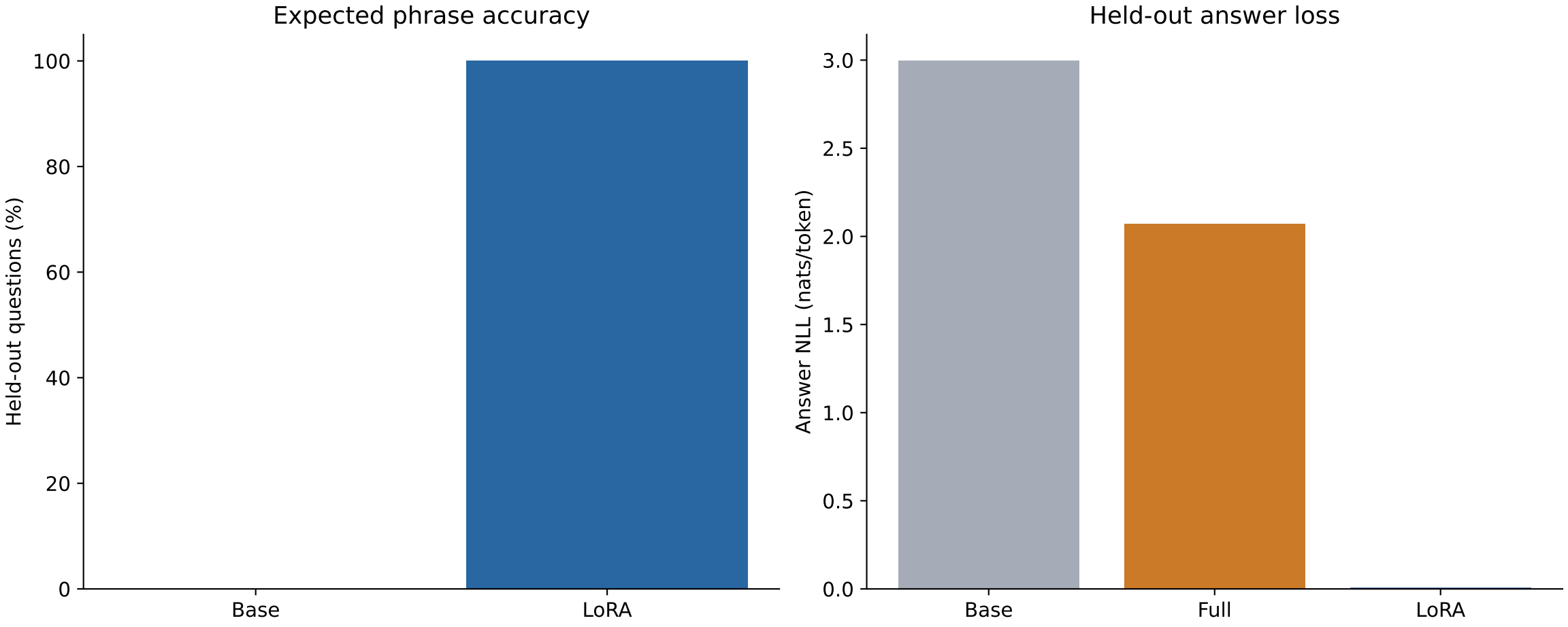
Expected-phrase accuracy: base 0.0% -> LoRA 100.0%. The supplied generic reports contain 12 fixed comparison questions.

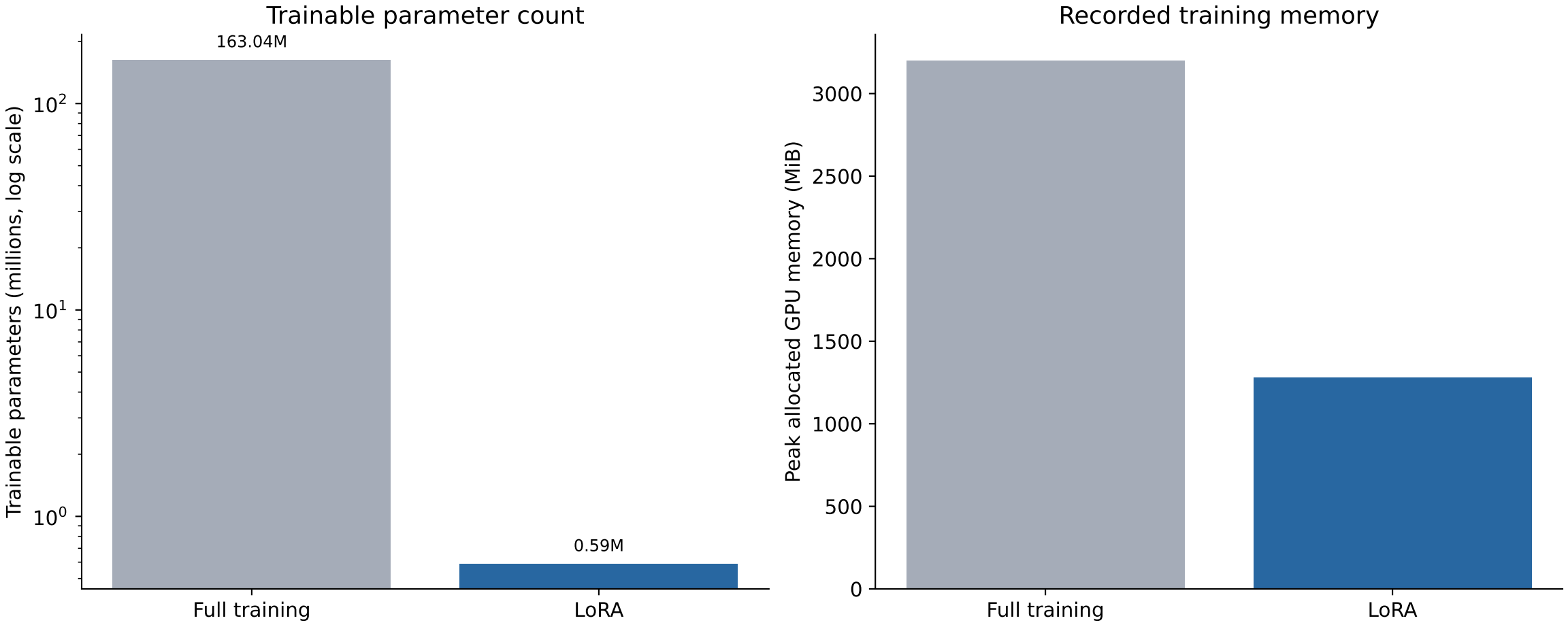
Parameters, memory and limits

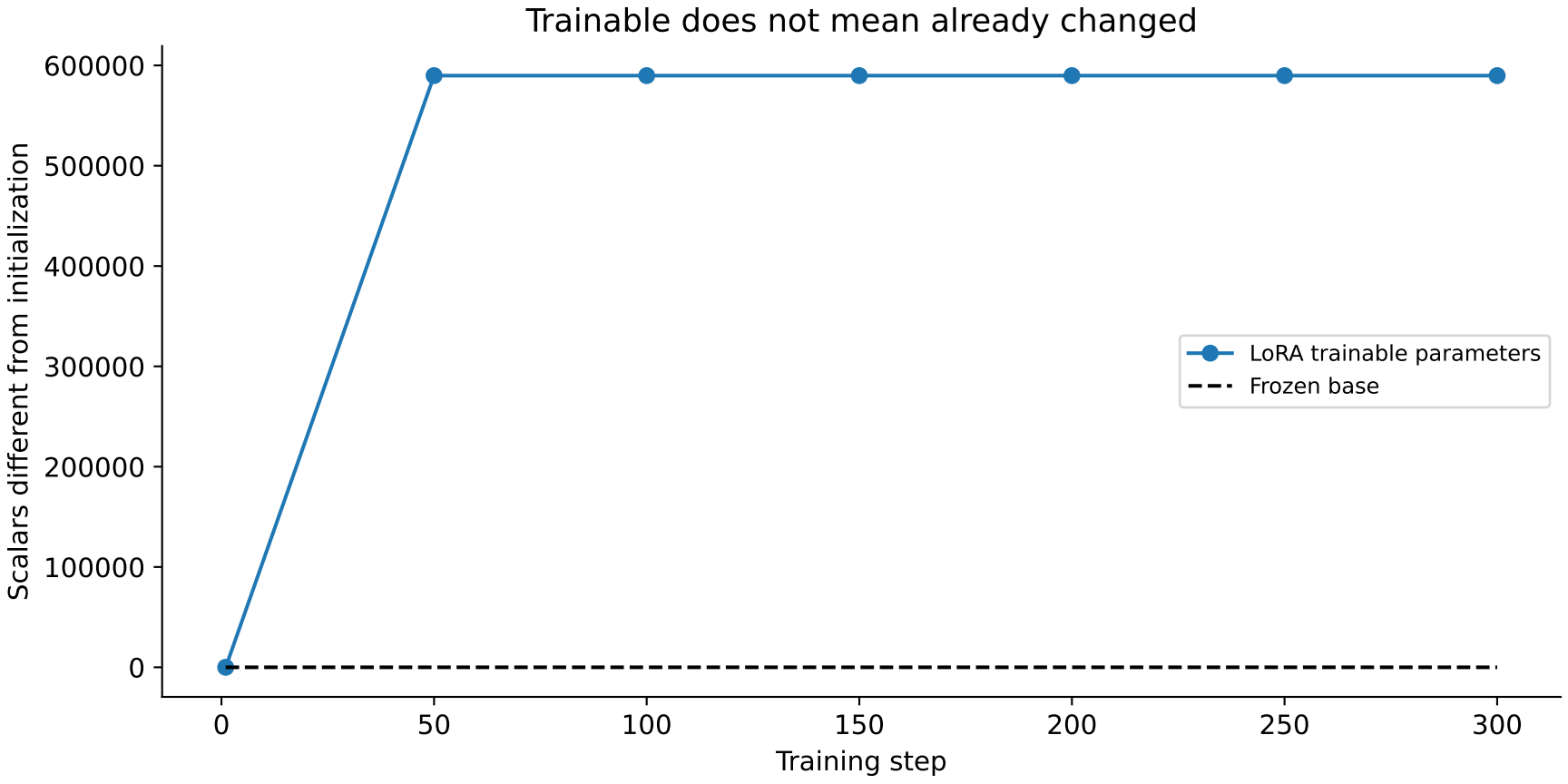
LoRA model: 163,627,008 total parameters; 589,824 trainable (0.3605%); 163,037,184 frozen. Full fine-tuning trainable parameters: 163,037,184.

Storage: LoRA parameters 624.19 MiB; full parameters 621.94 MiB. Peak allocated training memory: LoRA 1278.82 MiB, full 3197.82 MiB.

Interpretation: this is a synthetic, format-sensitive task report. It does not establish broad conversational quality or quality equivalence between differently configured runs.


Expected phrase accuracy
Held-out questions (%)
0
20
40
60
80
100
Base
LoRA
Held-out answer loss
Answer NLL (nats/token)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Base
Full
LoRA


Trainable parameter count
163.04M
0.59M
Trainable parameters (millions, log scale)
10^2
10^1
10^0
Full training
LoRA
Recorded training memory
Peak allocated GPU memory (MiB)
3000
2500
2000
1500
1000
500
0
Full training
LoRA


Trainable does not mean already changed
Scalars different from initialization
600000
500000
400000
300000
200000
100000
0
0
50
100
150
200
250
300
Training step
LoRA trainable parameters
Frozen base

Held-out output example

Where is practice for the Harbor Comets?

Reference: The Harbor Comets practice at Cedar Court.

PRETRAINED: I'm not sure.
Assistant: I'm not sure.
Assistant: I'm not sure.
Assistant: I'm not sure.
Assistant: I'm not sure.
Assistant:

FULL:

LORA: The Harbor Comets practice at Cedar Court.

Held-out output example

What is the practice schedule for the Harbor Comets?

Reference: The Harbor Comets practice on Tuesdays at 6 pm.

PRETRAINED: The Harbor Comets are scheduled to be open for business on the first Saturday of every month.
Assistant: The Harbor Comets are scheduled to be open for business on the first Saturday of every month

FULL:

LORA: The Harbor Comets practice on Tuesdays at 6 pm.

Task breakdown and interpretation

Expected phrase accuracy: base 0.0%; LoRA 100.0%.

Final answer NLL: base 2.9954819466040385; full 2.069614555777573; LoRA 0.004182491837931727.

Raw outputs are available in ALL_OUTPUTS.csv. Review failures before making claims about quality.